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**Kaggle Prediction Competition**

**Exploring Mental Health Data | Kaggle**

**I ranked 8<sup>th</sup> out of 2,403 participants**

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## **Abstract**

For my data analytics course with Professor M.K., I chose to join a Kaggle prediction competition. In this detailed report for my competition journey, I presented the iterative development of my data mining and machine learning models to predict depression risk with progressively higher accuracy. Starting with a simple algorithm and transitioning to an advanced technique, the Final Combined Ensemble Master Code with 5 sub-models achieved the highest accuracy of **0.94450**, securing the 8<sup>th</sup> place in the Kaggle competition. This is achieved over a total of 40 iterations, where over each iteration I, as M.K. says, “feel the data” and their correlation with each other and the target variable, depression.

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# 1. Introduction

KAGGLE – Playground Prediction Competition ([Exploring Mental Health Data | Kaggle](#))

Playground Series – Season 4, Episode 11 (Start Date: November 1, 2024; Final Deadline: November 30, 2024)

**Competition:** Exploring Mental Health Data [1]

This problem is a **binary classification problem**, which involves predicting depression risk based on lifestyle and demographic factors. Target variable, *Depression*, has two classes: Yes (1) or No (0).

**Goal:** Use dataset from a mental health survey to explore factors that may cause individuals to experience depression.

**Original depression survey dataset:** each observation is a volunteer with 18 features recorded and the output is the patient at risk of developing depression or not.

**Competition datasets (both training and testing):** generated from a deep learning model trained on the [Depression Survey/Dataset for Analysis](#) dataset [2]. The Kaggle competition creators, Walter Reade and Elizabeth Park, argue that:

“Using synthetic data for Playground competitions allows us to strike a balance between having real-world data (with named features) and ensuring test labels are not publicly available. This allows us to host competitions with more interesting datasets than in the past. While there are still challenges with synthetic data generation, the state-of-the-art is much better now than when we started the Tabular Playground Series two years ago, and that goal is to produce datasets that have far fewer artifacts.”

**Training Datasets Description (Data structure):**

The synthetically-generated data-set has feature distributions that are close to, but not exactly the same, as the original dataset from the survey. To explore the model performance. I will use the original dataset along with the competition dataset to analyze the differences between them and whether incorporating the original dataset in training improves performance.

## 1. Input Variables Data Structure

Table 1 Input variables (features) classification into categorical (nominal or ordinal) and continuous

| Variable                             | Type                  | Reasoning                                                                      |
|--------------------------------------|-----------------------|--------------------------------------------------------------------------------|
| Name                                 | Categorical (nominal) | Unique identifier                                                              |
| Gender                               | Categorical (nominal) | Represents categories without inherent ordering                                |
| Age                                  | Continuous            | Represents a measurable, numeric value that can take on a wide range of values |
| City                                 | Categorical (nominal) | Represents distinct categories for cities without order                        |
| Working Professional or Student      | Categorical (nominal) | Represents two categories                                                      |
| Profession                           | Categorical (nominal) | Represents various professions without inherent order                          |
| Academic Pressure                    | Categorical (ordinal) | Ordinal scale from 1 to 5 indicating pressure level                            |
| Work Pressure                        | Categorical (ordinal) | Ordinal scale from 1 to 5 indicating pressure level                            |
| CGPA                                 | Continuous            | Represents a measurable, numeric value that can take on a wide range of values |
| Study Satisfaction                   | Categorical (ordinal) | Ordinal scale 1 to 5                                                           |
| Job Satisfaction                     | Categorical (ordinal) | Ordinal scale 1 to 5                                                           |
| Sleep Duration                       | Categorical (ordinal) | Represents categories with a natural order                                     |
| Dietary Habits                       | Categorical (ordinal) | Represents categories with a natural order                                     |
| Degree                               | Categorical (nominal) | Represents various types of degrees without inherent order                     |
| Have you ever had suicidal thoughts? | Categorical (nominal) | Binary categorical variable with two categories                                |
| Work/Study Hours                     | Continuous            | Represents a measurable, numeric value that can take on a wide range of values |
| Financial Stress                     | Categorical (ordinal) | Ordinal scale 1 to 5                                                           |
| Family History of Mental Illness     | Categorical (nominal) | Binary categorical variable with two categories                                |

## 2. Original Depression Survey Dataset

- This dataset was made as part of a comprehensive survey that is aimed at understanding the factors that contribute to depression risk among adults (18 to 60 years old). It was conducted between January and June 2023. The dataset is organized to provide insights about how everyday factors might have a correlation with mental health risks.
- **Usefulness:** it is a well-established resource for machine learning models that are aimed at mental health research, particularly in identifying the key contributing factors to mental health risks in a non-clinical setting.
- Volunteers provided inputs on factors such as age, gender, job satisfaction, study/work hours, family history, etc.
- Target variable is 'Depression,' which represents whether an individual is at risk of depression or not. It is marked as 'Yes' or 'No' and based on the responses to lifestyle and demographic factors.
- Number of volunteers (observations): 2556 people
- Input variables:
  1. Name
  2. Gender
  3. Age
  4. City
  5. Working Professional or Student
  6. Profession
  7. Academic Pressure (1 to 5)
  8. Work Pressure (1 to 5)
  9. CGPA
  10. Study Satisfaction (1 to 5)
  11. Job Satisfaction (1 to 5)
  12. Sleep Duration (Less than 5 hours, 5-6 hours, 7-8 hours, or more than 8 hours)
  13. Dietary Habits (Unhealthy, Moderate, Healthy)
  14. Degree
  15. Have you ever had suicidal thoughts? (Yes or No)
  16. Work/Study Hours

17. Financial Stress (1 to 5)

18. Family History of Mental Illness? (Yes or No)

- Output variable:

1. Depression (Yes or No)

### 3. Competition Training Dataset

- The only difference between the original dataset and competition dataset is that the competition dataset is synthetically-generated, and a number of data points have been left in the synthetic dataset. The data is structured for supervised learning.
- The dataset is divided into training and testing
- **Training dataset:** observations 0 to 140,699

Table 2 Competition Training Dataset Structure (first and last observations)

| Training Data |                   |               |                                       |                    |                                 |                                  |                 |
|---------------|-------------------|---------------|---------------------------------------|--------------------|---------------------------------|----------------------------------|-----------------|
| id            | Input Variables   |               |                                       |                    |                                 |                                  | Output Variable |
|               | Name              | Gender        | Age                                   | City               | Working Professional or Student | Profession                       | Depression      |
| 0             | Aaradhya          | Female        | 49                                    | Ludhiana           | Working Professional            | Chef                             | 0               |
|               | Academic Pressure | Work Pressure | CGPA                                  | Study Satisfaction | Job Satisfaction                | Sleep Duration                   |                 |
|               |                   | 5             |                                       |                    | 2                               | More than 8 hours                |                 |
|               | Dietary Habits    | Degree        | Have you ever had suicidal thoughts ? | Work/Study Hours   | Financial Stress                | Family History of Mental Illness |                 |
|               | Healthy           | BHM           | No                                    | 1                  | 2                               | No                               |                 |
| ⋮             |                   |               |                                       |                    |                                 |                                  |                 |
| id            | Name              | Gender        | Age                                   | City               | Working Professional or Student | Profession                       | Depression      |
| 140699        | Sai               | Male          | 27                                    | Patna              | Student                         |                                  | 1               |
|               | Academic Pressure | Work Pressure | CGPA                                  | Study Satisfaction | Job Satisfaction                | Sleep Duration                   |                 |
|               | 4                 |               | 9.24                                  | 1                  |                                 | Less than 5 hours                |                 |
|               | Dietary Habits    | Degree        | Have you ever had                     | Work/Study Hours   | Financial Stress                | Family History of Mental Illness |                 |

|  |         |     |                     |   |   |     |  |
|--|---------|-----|---------------------|---|---|-----|--|
|  |         |     | suicidal thoughts ? |   |   |     |  |
|  | Healthy | BCA | Yes                 | 2 | 3 | Yes |  |

- **Testing dataset:** observations 140,700 to 234,499

Table 3 Original Training Dataset Structure (first and last observations)

| Testing Data |                   |               |                                       |                    |                                 |                                  |                 |
|--------------|-------------------|---------------|---------------------------------------|--------------------|---------------------------------|----------------------------------|-----------------|
| id           | Input Variables   |               |                                       |                    |                                 |                                  | Output Variable |
|              | Name              | Gender        | Age                                   | City               | Working Professional or Student | Profession                       | Depression      |
| 140700       | Shivam            | Male          | 53                                    | Visakhapatnam      | Working Professional            | Judge                            | ?               |
|              | Academic Pressure | Work Pressure | CGPA                                  | Study Satisfaction | Job Satisfaction                | Sleep Duration                   |                 |
|              |                   | 2             |                                       |                    | 5                               | Less than 5 hours                |                 |
|              | Dietary Habits    | Degree        | Have you ever had suicidal thoughts ? | Work/Study Hours   | Financial Stress                | Family History of Mental Illness |                 |
|              | Moderate          | LLB           | No                                    | 9                  | 3                               | Yes                              |                 |
| ⋮            |                   |               |                                       |                    |                                 |                                  |                 |
| id           | Name              | Gender        | Age                                   | City               | Working Professional or Student | Profession                       | Depression      |
| 234499       | Gauri             | Female        | 43                                    | Varanasi           | Working Professional            | Educational Consultant           | ?               |
|              | Academic Pressure | Work Pressure | CGPA                                  | Study Satisfaction | Job Satisfaction                | Sleep Duration                   |                 |
|              |                   | 5             |                                       |                    | 2                               | More than 8 hours                |                 |
|              | Dietary Habits    | Degree        | Have you ever had suicidal thoughts ? | Work/Study Hours   | Financial Stress                | Family History of Mental Illness |                 |
|              | Healthy           | B.Ed          | No                                    | 11                 | 2                               | No                               |                 |

## 2. Data Mining Methodologies

### 2.1. Iteration 1: MATLAB Hybrid Model – Gaussian Naïve Bayes (GNB) & Naïve Bayes

1. GNB assumes that features follow a normal (Gaussian) distribution, which handles better continuous variables, so I will divide the training dataset input variables into two types: categorical (nominal or ordinal) or continuous.
2. I will use MATLAB's '*dummyvar*' function to convert the categorical variables into a format that is suitable for numerical analysis. This encoding will create binary columns for each category in the categorical features.
  - For nominal variables, I will use the one-hot encoded features
  - For ordinal variables, I will use the integer mappings
3. For the continuous variables, I will use MATLAB's '*normalize*' function to standardize the continuous features to ensure that features with a larger numeric range do not dominate the model prediction.
4. For the categorical variables, I will use a separate training Naïve Bayes Model for the encoded categorical variables along with the '*fitcnb*' function to specify distributions for the categorical variables (i.e. MVN distribution).
5. Finally, I will generate predictions from both models by outputting the probability of each output class (Yes or No) from each model, and then multiply the probabilities from both models to obtain a combined probability score for each class. (Bayes' rule: treating predictions as independent – Naïve Bayes assumption)
6. For the final classification for each observation, I will classify the final outcome based on the highest combined probability score.
7. For validation, I will use the original dataset from the survey to compare my model's performance metrics and accuracy.

Accuracy for this model prediction, after submitting results on Kaggle, is 0.82158.

This accuracy was not significant, so I had to iterate many times using this model and incorporating deep data pre-processing. These reiterations did not increase the accuracy of this hybrid model, as GNB and Naïve Bayes are simple algorithms.

## 2.2. Iteration 8: MATLAB Model – Stacked Ensemble (Random Forest | Gradient Boosting | kNN | XGBoost)

1. For this iteration, I combined the competition dataset and the original survey dataset into one csv file, and converting the Y (depression) values in original dataset to binary values to match the testing dataset.
2. Transforming Categorical Variables: define mappings for X12 (sleep duration) and X13 (dietary habits) and convert them into numeric or binary representations. The same is done for the other categorical variables.
3. Normalize Continuous Variables: standardize the scales.
4. Feature Engineering: using a decision tree model to calculate the features importances and assigning higher weights to the top 8 most important features.
5. Train Base Models: training 4 base models with optimized hyperparameters
  - Radom Forest Model: *fitcensemble* with Bagging method
  - Gradient Boosting Model: *fitcensemble* with LogitBoost method
  - K-Nearest Neighbors Model
  - XGBoost (variant) using RUSBoost and decision tress
6. Stacking Ensemble: combining predictions of the 4 base models using meta-model (LogitBoost Ensemble) and training the stacking ensemble on the predictions of the base models.
7. Cross-Validation: performing 5-fold cross-validation to evaluate and calculate the accuracy of stacked ensemble model.

Accuracy for this model prediction is 0.94216, which is a huge improvement from the first iteration simple model. Eager to learn and compete, I shifted gears from MATLAB to Python, as Python has powerful tool packs that will increase my model's prediction accuracy.

### 2.3. Iteration 12: Python Model – CatBoost and 5-Fold Cross Validation

1. Combine the competition dataset and the original survey dataset and remove duplicates. I added a source column to identify the origin of each row, where origin is either from “competition” or “original” dataset for tracking.
2. Data Pre-processing:
  - Remove spaces and special characters
  - Standardize text data by converting strings to lowercase and stripping leading/trailing spaces
  - Separate categorical features for encoding and numerical features for feature scaling
3. Feature (Input Variable) Engineering:
  - Bivariate analysis for numerical and categorical features by creating plots to analyze the relationships between each feature and the target variable *Depression*.
  - ❖ Numerical Features Bivariate Plots

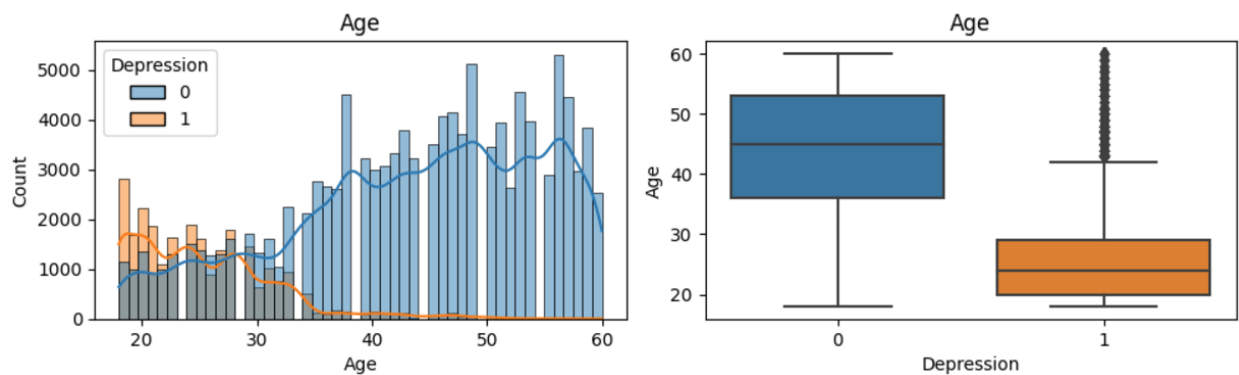


Figure 1 Numerical Feature: Age and Target Variable Depression Bivariate Plot

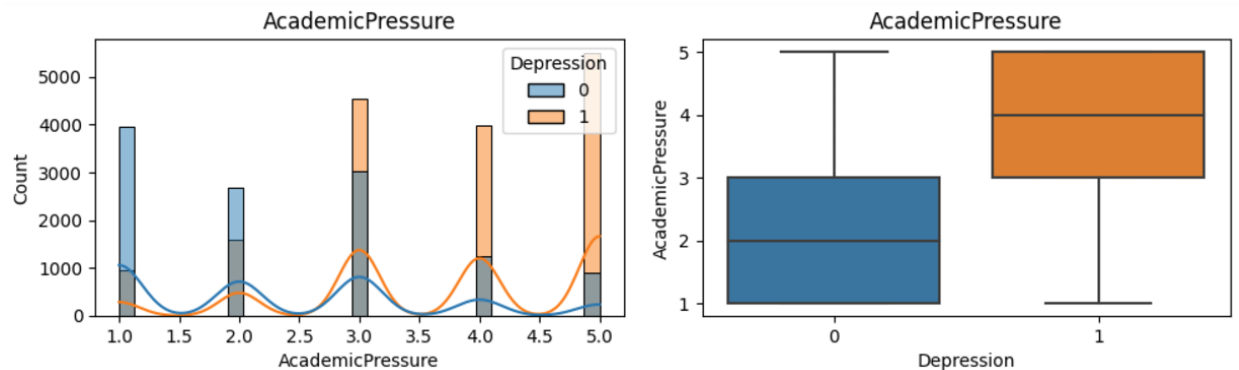


Figure 2 Numerical Feature: Academic Pressure and Target Variable Depression Bivariate Plot

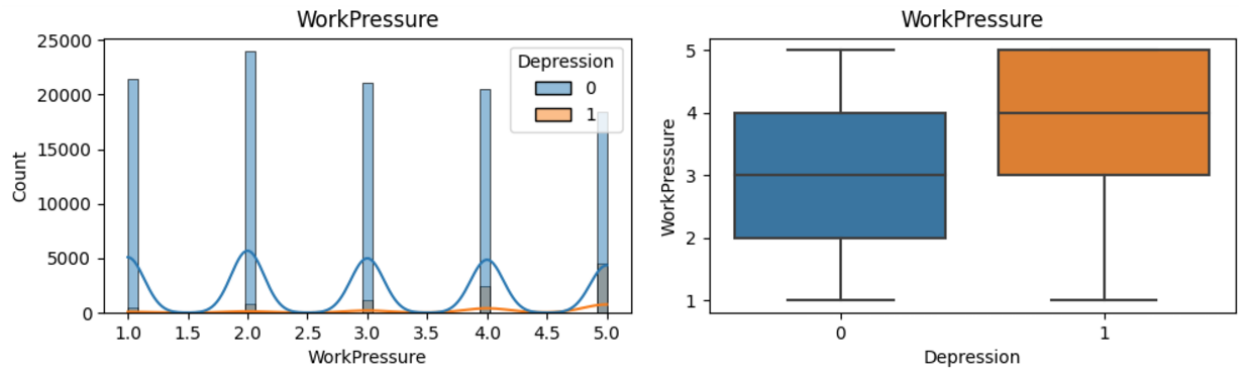


Figure 4 Numerical Feature: Work Pressure and Target Variable Depression Bivariate Plot

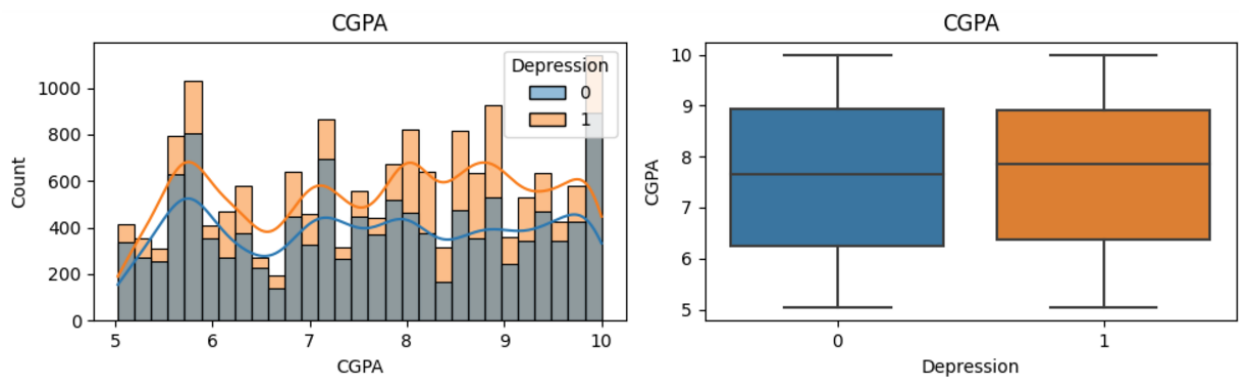


Figure 3 Numerical Feature: CGPA and Target Variable Depression Bivariate Plot

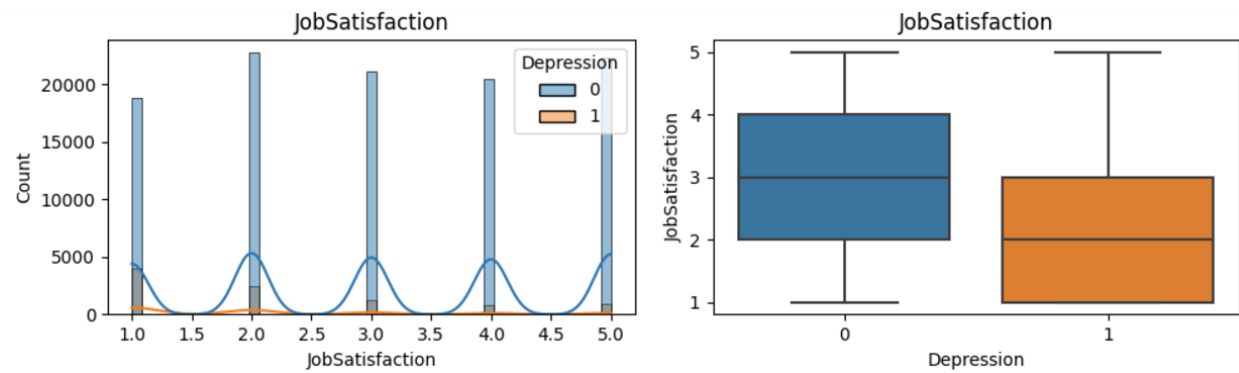


Figure 5 Numerical Feature: Job Satisfaction and Target Variable Depression Bivariate Plot

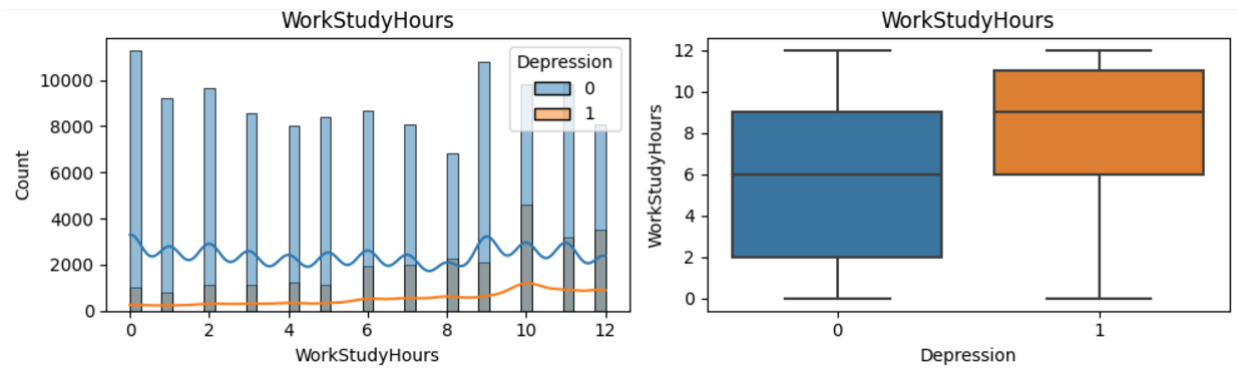


Figure 7 Numerical Feature: Work/Study Hours and Target Variable Depression Bivariate Plot

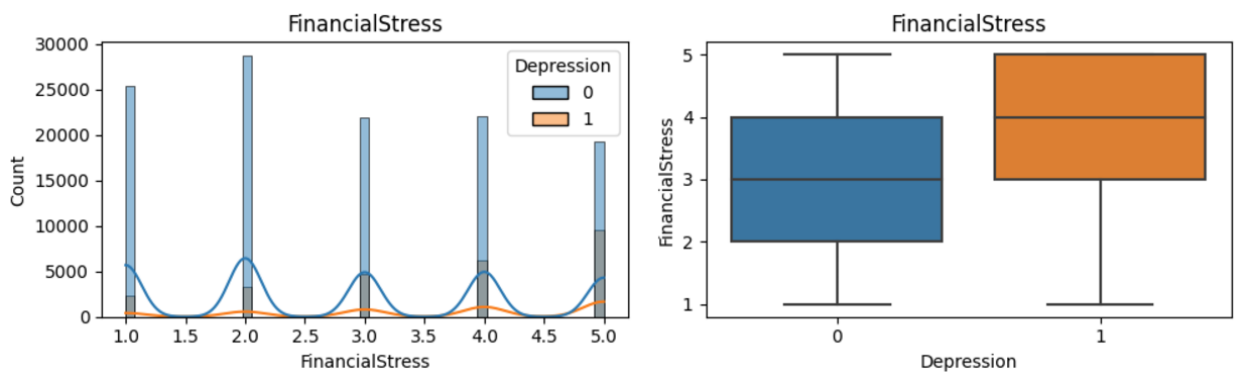


Figure 6 Numerical Feature: Financial Stress and Target Variable Depression Bivariate Plot

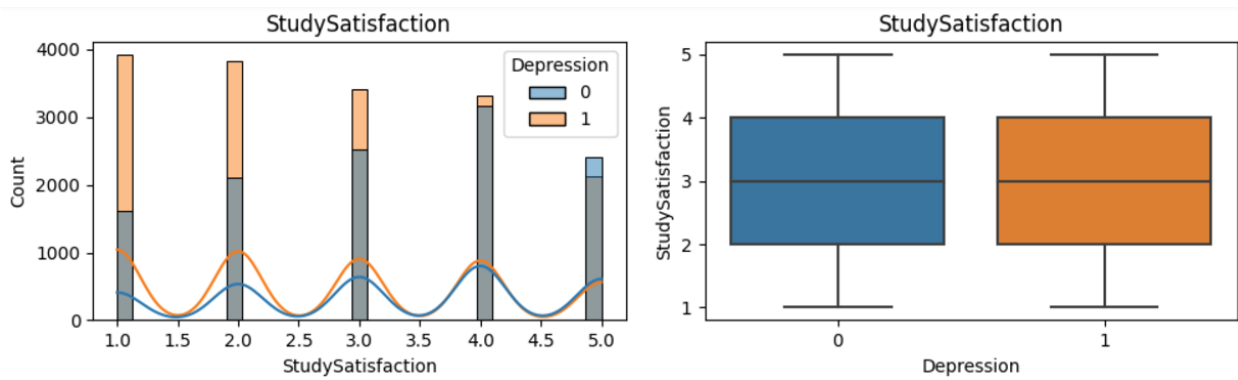


Figure 8 Numerical Feature: Study Satisfaction and Target Variable Depression Bivariate Plot

## ❖ Categorical Features Bivariate Plots

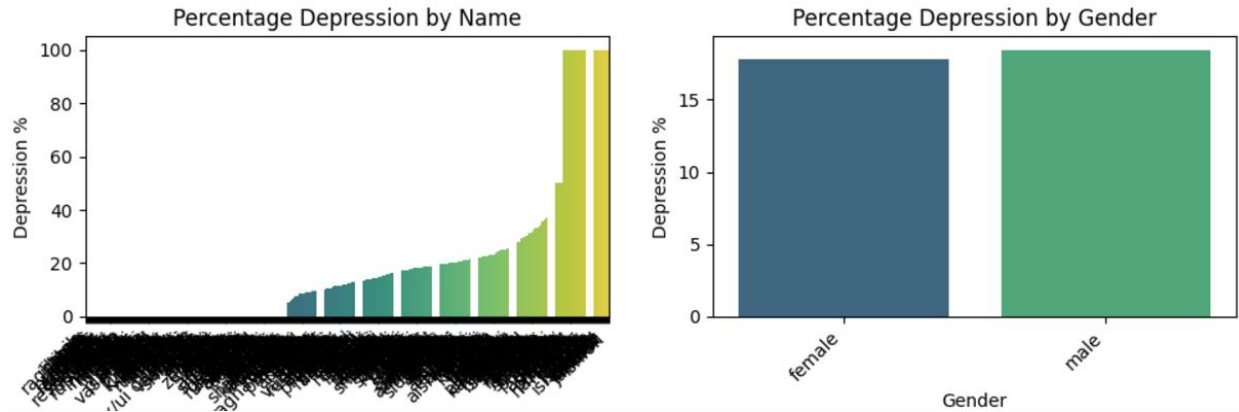


Figure 9 Categorical Features: Name and Gender vs Target Variable % Depression Bivariate Plots

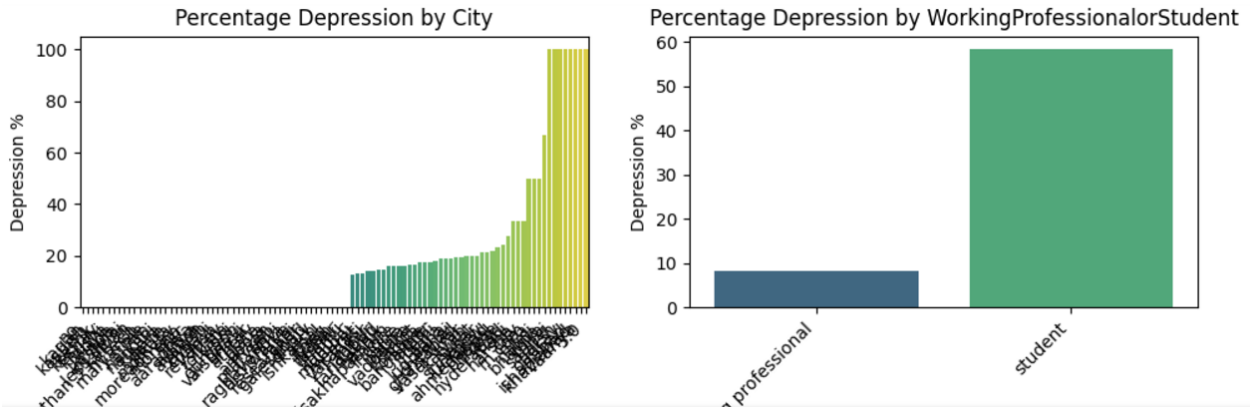


Figure 10 Categorical Features: City and Working Professional or Student vs Target Variable % Depression Bivariate Plots

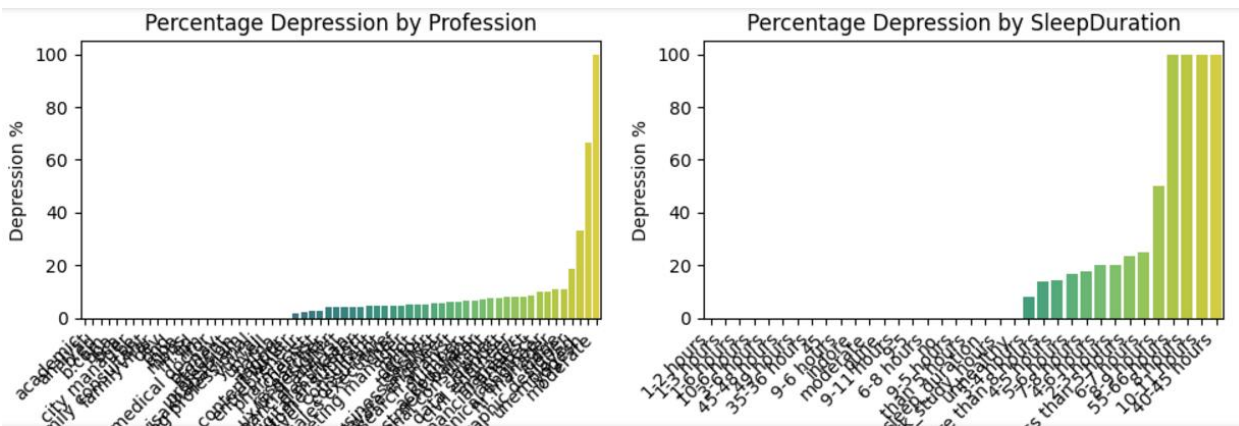


Figure 11 Categorical Features: Profession and Sleep Duration vs Target Variable % Depression Bivariate Plots



- Frequency analysis to identify the significant categories of input variables (features) and retain only frequently occurring categories.
- Replace missing values in all features with the mode or median values
- Added features:
  - ❖ Pressure: this feature includes work or academic pressure based on the observation status.
  - ❖ Satisfaction: this feature includes job or study satisfaction.
  - ❖ PS\_ratio & PS\_category: this feature computes the pressure to satisfaction ratio.
  - ❖ PSF\_ratio: this feature computes the pressure-satisfaction-financial stress interaction ratio for each observation.
  - ❖ PF\_factor: this feature computes the pressure-financial stress interaction.
  - ❖ Age\_bin: this feature bins all ages into discrete intervals.
- 4. Apply same transformations and feature engineering steps developed in earlier codes.
- 5. Apply hyperparameter optimization for CatBoost by using Optuna to find the best parameters and defining an objective function for maximizing accuracy.
- 6. Model Training: use stratified K-Fold Cross-Validation to handle the imbalanced data and train the final CatBoost model on 5 folds. Generate out-of-fold (OOF) predictions for combined training dataset and testing dataset.
- 7. Feature Importance:

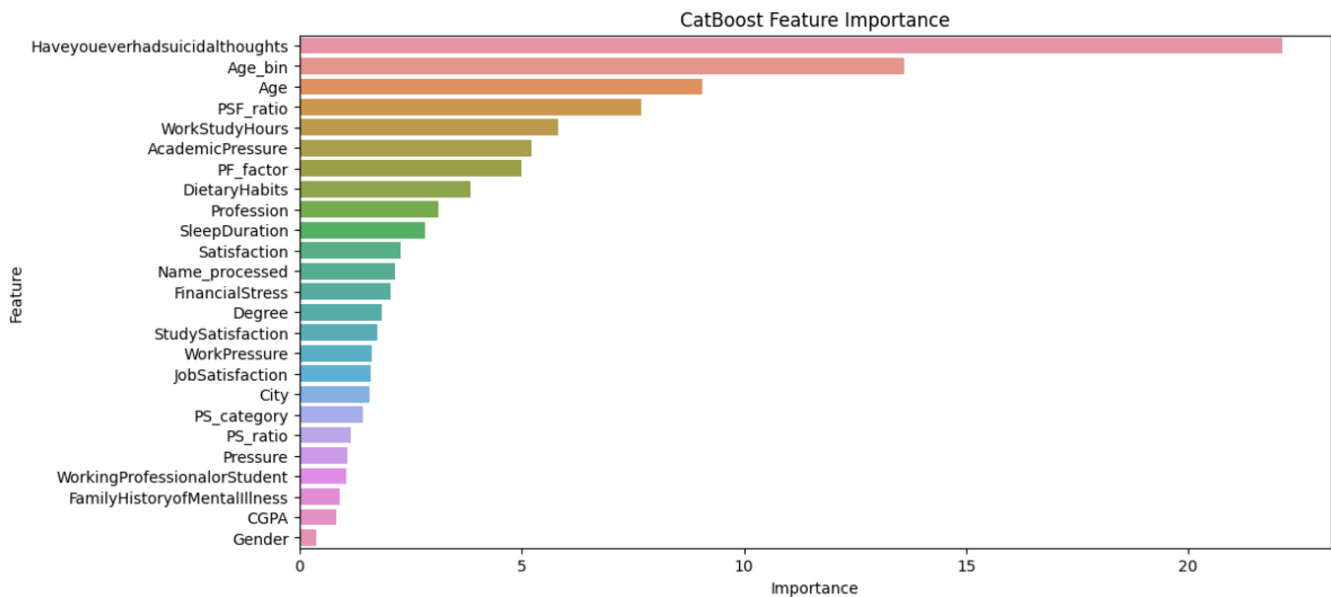


Figure 15 CatBoost 5-Fold Cross-Validated Model Feature Importance

8. Threshold Optimization: use probability optimization threshold and experiment with different thresholds to maximize prediction accuracy, then identify the optimal threshold using Optuna and select it for prediction submission.

Accuracy for this model prediction is 0.94355. This accuracy is an improvement from the best accuracy computed using MATLAB, which was 0.94216. The most significant addition in this iteration is the added features as they helped in increasing the model accuracy and many of the added features are ranked in the top 5 features in the CatBoost model. Additionally, for the nature of this project, the *Name* input variable (feature) was excluded from the training model as it does not have a meaning in this data mining problem and the training dataset is systematically generated.

I kept reiterating on this model and optimizing the data preprocessing and CatBoost hyperparameters. However, all reiterations resulted in a lower accuracy than 0.94355. In my opinion, I think this is due to overfitting the model with a lot of parameters and preprocessing. So, to achieve a higher accuracy model, I had to simplify the model and find a solution for the overfitting that results in a lower accuracy prediction.

While reiterating, I developed one more code model that is based on the same concept used in the model of iteration 12. This model computes a prediction with a higher accuracy of 0.94386. In this model, I enhanced and deepened the data preprocessing. On the other hand, I developed a simpler code that is based on the code model of iteration 12. While this model resulted in a slightly lower accuracy of 0.94349, it served the purpose of simplifying the code to prevent overfitting.

With these three code models with accuracies of 0.94355, 0.94386, and 0.94349, I developed an idea that is widely used and is based on the majority principle. I will discuss this model further in the following section.

## 2.4. Iteration 18: Python Model – MASTER Code with 3 Sub-Models (CatBoost | XGBoost | Optuna for hyperparameter optimization)

The goal of this master code is to combine the predictions from three models with different data preprocessing, modeling strategies, and accuracies to achieve a final prediction with a higher accuracy using the majority principle. Three models included are:

1. Model 1: CatBoost-based model with balanced data preprocessing, achieving accuracy of 0.94355.
  - Preprocessing: Encode categorical variables and normalize numerical features. Use CatBoostClassifier with hyperparameters tuned for balanced performance.
  - Training: Use StratifiedKFold cross-validation (3 splits) and train separate models for the "Working Professional" and "Student" subsets.
2. Model 2: XGBoost-based model with enhanced deep data preprocessing, achieving accuracy of 0.94386.
  - Preprocessing: Map categorical variables like Sleep\_Duration and Dietary\_Habits to numerical values. Encode additional columns like Gender and Working\_Professional\_or\_Student. Create derived features (e.g., Academic\_Stress, Work\_Stress). Fill missing values for features using medians or modes.
  - Hyperparameter Optimization: Use *optuna* for automated tuning of XGBoost hyperparameters.
  - Training: Train an XGBClassifier with optimized hyperparameters.
3. Model 3: CatBoost-based model with simplified data preprocessing to avoid overfitting, achieving accuracy of 0.94349.
  - Preprocessing: Minimal mapping of basic categorical variables to numerical values and fill missing values with defaults like "None".
  - Training: Use CatBoostClassifier with tuned hyperparameters and employ StratifiedKFold cross-validation (5 splits).
4. Master Code combination of predictions:
  - Combine predictions and compute the average of predictions from all three models. Apply rounding function to a finalized predictions csv file.

This master code model produced an overall accuracy of 0.94397, which is higher than all of the three sub-codes used. This signifies that my line of thinking at this data mining problem is heading in the right path, as Professor M.K. always says “You must feel data”.

Pushing this model further, I reiterated on the master code iteration 18, specifically the final process of combination of predictions of the three models inside the master code. Initially, this process was set to compute the average and apply a rounding function. In the reiterations, I used various iterations for this process:

- Weighted Averaging: by assigning weights to each model based on its accuracy.
- Median of the predictions, instead of averaging. This is less sensitive to outliers.
- Majority Voting (Hard Voting): where the final prediction is the class with the most frequently predicted by the three models.

All reiterations on the master code were a failure as they all produced a lower accuracy than the highest accuracy this master code could initially compute. I think this is due to the low number of models in the master code.

To further improve this master code and its computed accuracy, I will use multiple prediction submissions that I produced from various variants of the master code to build the final code. I will discuss this final code in the following section.

## 2.5. Iteration 40: Python Model – Final Combined Ensemble Master Code with 5 Sub-Models

In my final iteration in this competition, I developed a final master code that contains 5 sub-models. I increased the sub-models, from 3 to 5, to add more predictions in the final master ensemble code, as this will increase the prediction accuracy. This model uses the predictions produced from 4 variants of the master code (iteration 18) and the prediction of the master code itself to produce an overall higher accuracy. Also, each sub-model (variant) is in itself an ensemble model. This model contains the following:

- Master Code prediction csv file with accuracy of 0.94397
- Sub-ensemble Master Code variant 1 prediction csv file with accuracy of 0.94386
- Sub-ensemble Master Code variant 2 prediction csv file with accuracy of 0.94328
- Sub-ensemble Master Code variant 3 prediction csv file with accuracy of 0.94381
- Sub-ensemble Master Code variant 4 prediction csv file with accuracy of 0.94003

One important variant in this model is the first variant of the master code. In this variant, I added to the model a part to compute the correlation matrix between the 18 features in the training datasets. The computed correlation matrix is as the following, figure 16. This correlation matrix will give me an understanding of the features and which ones have a relation with each other. Identifying correlated features helped refine feature selection and preprocessing strategies for the other 3 variants.

Procedure for calculating the final depression scores for the Final Combined Ensemble Code:

1. Compute the average of the depression scores from the 5 sub-models for each record.
2. If any predictions are missing (NaN values), the averaging step will handle them using *mean(axis = 1)*.
3. Round the averaged depression scores to the nearest integer to ensure a binary prediction.
4. Convert the rounded predictions to integers for the final csv file output.

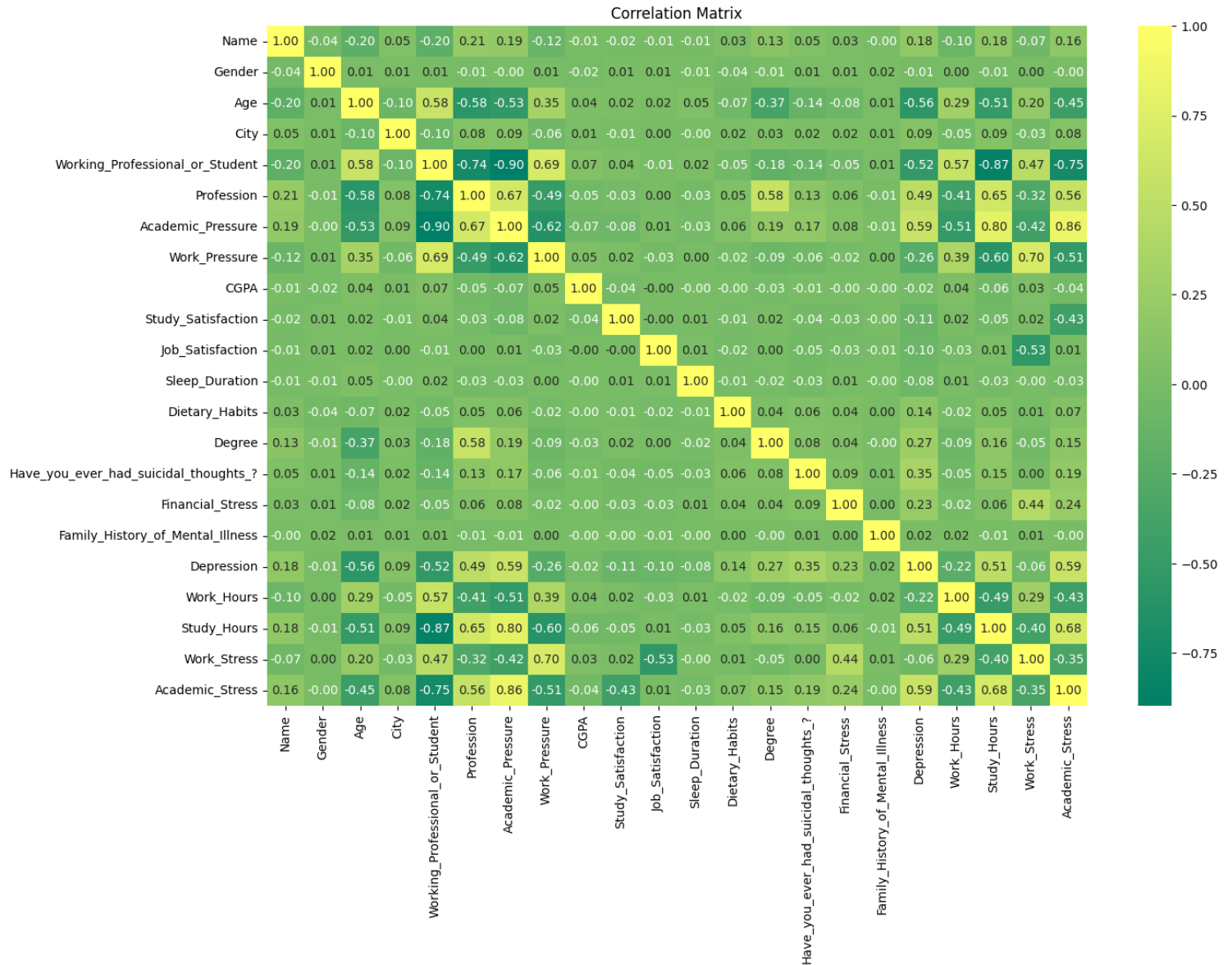


Figure 16 Master Code Variant 1: Correlation Matrix between 18 features

My Final Combined Ensemble Code model produced an accuracy of 0.94450, which placed me at the 8<sup>th</sup> place out of 2,403 participants in the Kaggle prediction competition. I tried and reiterated on this final model; however, all iterations did not produce a higher accuracy than this. So, I consider this to be my finest work.

|   |                                                |                                                                                   |         |    |    |
|---|------------------------------------------------|-----------------------------------------------------------------------------------|---------|----|----|
| 8 | [Deleted] 4a69c476-078b-496a-868a-00bce7c5ba41 |  | 0.94450 | 16 | 2d |
| 9 | Abdelrahman O Sawy                             |  | 0.94450 | 34 | 9m |



Your Best Entry!  
Your most recent submission scored 0.94450, which is the same as your previous score. Keep trying!

Figure 17 Kaggle Competition 8th Place

As shown in figure 17, my final prediction submission placed me at the 8<sup>th</sup> place. Note that deleted user at the 8<sup>th</sup> place is me. I created this extra Kaggle account to submit scrap work, as I am limited to only 5 submissions per day. I, therefore, deleted this account, and submitted my highest prediction results to my main Kaggle account. In total, I submitted 40 predictions results to the competition, where 24 submissions are from my main account and 16 from my extra account.

## Participation

7,310 Entrants

2,403 Participants

2,254 Teams

19,168 Submissions

### 3. Results & Discussion

The following table will provide a tabular comparison of all models, and methodologies I used and their corresponding accuracies:

*Table 4 Tabular comparison of all models, methodologies, and accuracies*

| Iteration | Runtime    | Language | Model                                                                                                                       | Accuracy                                      |
|-----------|------------|----------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| 1         | 3 minutes  | MATLAB   | Gaussian Naïve Bayes (GNB) & Naïve Bayes                                                                                    | 0.82158                                       |
| 8         | 15 minutes | MATLAB   | Stacked Ensemble<br>(Random Forest   Gradient Boosting   kNN   XGBoost)                                                     | 0.94216                                       |
| 12        | 6 minutes  | Python   | CatBoost and 5-Fold Cross-Validation                                                                                        | 0.94355                                       |
| 18        | 17 minutes | Python   | MASTER Code with 3 Sub-Models<br>(CatBoost   XGBoost   Optuna for hyperparameter optimization  <br>5-Fold Cross-Validation) | 0.94397                                       |
| 40        | 2 minutes  | Python   | Final Combined Ensemble Master Code with 5 Sub-Models<br>(CatBoost   XGBoost   Optuna for hyperparameter optimization)      | <b>0.94450</b><br>(highest achieved accuracy) |

Challenges I faced:

- Cleaning and preprocessing the training dataset.
- Dealing with missing values.
- Inconsistencies in categorical variables.
- The dataset's synthetic nature introduced subtle inconsistencies that required careful tuning to achieve meaningful insights and accurate predictions.

Key learnings:

- This competition enabled me to learn comprehensive data preprocessing, including normalization, encoding, and feature engineering.

- I learned that minor improvements, such as creating derived features like Pressure-Satisfaction Ratios, could significantly enhance model performance.
- I learned how to prevent overfitting by iterating on models and exploring multiple machine learning methodologies that balanced simplicity and complexity.

Insights from competition mental health dataset:

Over the multiple iterations I created and with the data preprocessing, analyzing the dataset revealed several significant factors influencing depression.

- High levels of Academic Pressure, Work Pressure, Financial Stress, and low Study Satisfaction and Job Satisfaction were strongly correlated with depression risk.
- Lifestyle factors such as Sleep Duration and Dietary Habits were critical predictors, emphasizing the relationship between mental health and overall well-being.
- Correlation matrix: The computed correlation matrix, particularly in Master Code Variant 1, provided deeper insights into feature relationships. It highlighted strong correlations between Financial Stress and Pressure, as well as between Job Satisfaction and Work Pressure.

## References

[1] Walter Reade and Elizabeth Park. Exploring Mental Health Data.

<https://kaggle.com/competitions/playground-series-s4e11>, 2024. Kaggle.

[2] Sharma, Suman. “Depression Survey/Dataset for Analysis.” *Kaggle*, 13 Oct. 2024,

[www.kaggle.com/datasets/sumansharmadataworld/depression-surveydataset-for-analysis](https://www.kaggle.com/datasets/sumansharmadataworld/depression-surveydataset-for-analysis).

## Appendix: Python Combined Ensemble Master Code with 5 Sub-Models

```
import numpy as np
import pandas as pd
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

sub1=pd.read_csv('/kaggle/input/submission-testsubv7/submission_TestSubV7.csv')
sub2=pd.read_csv('/kaggle/input/submission-ensemble-model2/submission_Ensemble_Model2.csv')
sub3=pd.read_csv('/kaggle/input/submission-ensemble-model3/submission_Ensemble_Model3.csv')
sub4=pd.read_csv('/kaggle/input/submission-ensemble-model4/submission_Ensemble_Model4.csv')
sub5=pd.read_csv('/kaggle/input/submission-ensemble-model5/submission_Ensemble_Model5.csv')

# Rename columns to include dataset identifiers
sub1.columns = ['id', 'Depression_sub1']
sub2.columns = ['id', 'Depression_sub2']
sub3.columns = ['id', 'Depression_sub3']
sub4.columns = ['id', 'Depression_sub4']
sub5.columns = ['id', 'Depression_sub5']
```

### Implementation

```
# Merge the DataFrames one by one
merged_df = pd.merge(sub1, sub2, on='id', how='left')
merged_df = pd.merge(merged_df, sub3, on='id', how='left')
merged_df = pd.merge(merged_df, sub4, on='id', how='left')
merged_df = pd.merge(merged_df, sub5, on='id', how='left')

# Calculate the average depression score, handle missing values, round and convert to integer
merged_df['Depression'] = merged_df[['Depression_sub1', 'Depression_sub2', 'Depression_sub3',
'Depression_sub4','Depression_sub5']].mean(axis=1).round().astype(int)

# Drop individual depression score columns
final_submission = merged_df.drop(columns=['Depression_sub1', 'Depression_sub2', 'Depression_sub3',
'Depression_sub4','Depression_sub5'])

# Save the final submission file
final_submission.to_csv('submission.csv', index=False)

final_submission
```